

Labour Demand, Nonwage Benefits, and Quasi-Fixed Costs

ECON 300 - Labour Economics

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- Outline quasi-fixed costs of labour and how they differ from variable (per-hour) costs.
- Distinguish workers vs. hours as separate margins of labour input.
- Explain the role of nonwage benefits in total compensation and in labour-demand modelling.
- Show how quasi-fixed costs create a wedge between VMP and the wage, shaping hoarding and dynamics.

Learning Objectives

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- Why are less-expensive, unskilled workers more likely to be laid off in downturns?
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Figure 6.1: Components of Total Compensation — *Placeholder*

Figure 6.1 — Components of total compensation.

Insert figure/diagram here later.

Figure 6.2: Total Compensation, Wage, and Nonwage Benefits — *Placeholder*

Figure 6.2 — Wage vs. nonwage benefits over time/levels.

Insert figure/diagram here later.

Unconditional Cash Transfers vs. Transfers-In-Kind

Employee Benefits

- Taxes
- Group purchase
- Adverse selection

Employer Benefits

- Production planning
- Deferred compensation

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- Arise from:
 - Hiring costs
 - Training costs
 - Dismissal costs
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Variable vs. Quasi-Fixed Costs

- **Variable** labour costs: vary with hours.
- **Quasi-fixed**: incurred per employee, *independent* of hours.

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Recurring vs. Nonrecurring

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General Effect of Quasi-Fixed Costs

- Raise MC of hiring another **worker** relative to adding **hours** on existing workers.
- Discourage expansion along the **employment** margin; encourage more hours per worker.
- Hiring continues until present value of expected future revenues equals present value of additional costs.

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Profit-Maximizing Employment Rule

$$\underbrace{H+T}_{\text{hiring+training}} + \sum_{t=1}^N \frac{W_t}{(1+R)^t} \leq \sum_{t=1}^N \frac{\text{VMP}_t}{(1+R)^t}$$

Where: $H + T$ = hiring and training costs; W_t = wage in period t ; R = discount rate; N = expected length of employment; $\text{MRP}_t(\text{VMP}_t)$ = expected value of marginal product.

WORKED EXAMPLE: Quasi-fixed Costs and Training

Costs	Employment Duration (N)	Fixed (H + T)	Variable ($\sum W_t$)	Total	Benefits ($\sum VMP_t$)	Total
	1 year	\$15,000	\$25,000	\$40,000	\$30,000	-\$10,000
	2 years	\$15,000	\$50,000	\$65,000	\$60,000	-\$5,000
	3 years	\$15,000	\$75,000	\$90,000	\$90,000	\$0
	4 years	\$15,000	\$100,000	\$115,000	\$120,000	\$5,000
	5 years	\$15,000	\$125,000	\$140,000	\$150,000	\$10,000

Some Phenomena Explained by Fixed Costs

- Overtime
- Temporary help agencies
- Layoffs
- Segmentation of labour markets
- Resistance to work-sharing

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Figure 6.3: Nonrecurring Fixed Employment Costs and Changes in Labour Demand — *Placeholder*

Figure 6.3 — Labour-demand shifts with nonrecurring fixed employment costs.

Insert figure/diagram here later.

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- Nonwage Benefit and Part-time work
- Short-time Work and the Great Recession of 2008–2009
- Subsidizing Employment Sharing
- Workers and the Demand for Worksharing

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- Quasi-fixed costs matter for employment vs. hours choices.
- These costs create a **wedge** between VMP and the wage.
- Implications for overtime, layoffs, segmentation, and worksharing policies.

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Thank You!